

# JSON Exercises

Complete the following exercises. Questions 1-2 use HTML & JavaScript, while Questions 3-4 use React.

## Question 1 – Student Report (HTML + JavaScript) ■■

A school stores student information in JSON format.

Example:

```
{id:1,name:"John",age:15,score:85}
```

Task:

- Display all students in an HTML table.
- Show ID, Name, Age and Score.
- Below the table display Highest Score, Lowest Score, Average Score and Number of Students.

Bonus:

Highlight students whose score is 90 or above.

## Question 2 – Pokémon Encyclopedia (HTML + JavaScript) ■■■

A Pokémon website stores Pokémon data in JSON.

Task:

- Display every Pokémon as a card.
- Each card shows Name, Type and Level.
- Add Search by Name.
- Filter by Type.
- Sort by Level.

Bonus:

Display Average Level and Highest Level Pokémon.

## Question 3 – Movie Collection (React) ■■■

A React application receives movie data from a JSON file.

Task:

- Store the JSON in state.
- Use map() to display movies as cards.
- Display Title, Genre and Rating.
- Add Search by Title.
- Filter by Genre.
- Sort by Rating.

Bonus:

Display Average Rating and Highest Rated Movie.

## Question 4 – Online Store Dashboard (React) ■■■■

A shop receives product information from a JSON file.

Task:

- Display products in a table.
- Show Name, Category, Price and Stock.
- Add Search by Product Name.
- Filter by Category.
- Sort by Price and Stock.

Below the table display:

- Total Products
- Average Price
- Most Expensive Product
- Total Stock

Bonus:

Highlight products with stock below 5 or display a 'Low Stock' label.

## Learning Objectives

- Read JSON objects and arrays.
- Display JSON data using HTML and React.
- Use map() to render lists.
- Search, filter and sort JSON data.
- Calculate statistics from JSON data.